Tumasulo with Reorder Example 2

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ROB Example Cycles 44-48 - Georgia Tech - HPCA: Part 3

https://www.youtube.com/watch?v=39AFF5Qq5DI&list=WL&index=12&t=13s

https://www.youtube.com/watch?v=V0nywwV0lKU

Sr	Inst	Operands

Architecture

Add= 1 Cycle
Mult= 10 Cycle
Div= 40 Cycle

RS Fields	Op	Dst-Tag	Tag 1	Tag 2	Val 1	Val 2

ROB Fields	Type	Dst	Value	Done	

RS Adder

RS Mul / Div

Cycle #

ARF

RAT

ROB

	Is	Ex	Wr	Cm

Sr	Inst	Operands
1	DIV	R2,R3,R4
2	MUL	R1,R5,R6
3	ADD	R3,R7,R8
4	MUL	R1,R1,R3
5	SUB	R4,R1,R5
6	ADD	R1,R4,R2

Add= 1 Cycle
Mult= 10 Cycle
Div= 40 Cycle

RS Fields

Op	Dst-Tag	Tag 1	Tag 2	Val 1	Val 2

ROB Fields

Type	Dst	Value	Done	

RS Mul / Div

DIV	ROB1				45	5

RS Adder

Cycle # 1

R1	-23
R2	16
R3	45
R4	5
R5	3
R6	4
R7	1
R8	2

ARF

R1	
R2	ROB1
R3	
R4	
R5	
R6	
R7	
R8	

RAT

ROB

ROB1	DIV	R2		
ROB2				
ROB3				
ROB4				
ROB5				
ROB6				

	Is	Ex	Wr	Cm
1	1	2		
2				
3				
4				
5				
6				

Sr	Inst	Operands
1	DIV	R2,R3,R4
2	MUL	R1,R5,R6
3	ADD	R3,R7,R8
4	MUL	R1,R1,R3
5	SUB	R4,R1,R5
6	ADD	R1,R4,R2

Add= 1 Cycle
Mult= 10 Cycle
Div= 40 Cycle

RS Fields

Op	Dst-Tag	Tag 1	Tag 2	Val 1	Val 2

ROB Fields

Type	Dst	Value	Done	

RS Mul / Div

DIV	ROB1				45	5

RS Adder

Cycle # 2 Continued next page

R1	-23
R2	16
R3	45
R4	5
R5	3
R6	4
R7	1
R8	2

ARF

R1	
R2	ROB1
R3	
R4	
R5	
R6	
R7	
R8	

RAT

FUTURE VAL

ROB

ROB1	DIV	R2	45/5=9	
ROB2				
ROB3				
ROB4				
ROB5				
ROB6				

	Is	Ex	Wr	Cm
1	1	2	42	
2				
3				
4				
5				
6				

Sr	Inst	Operands
1	DIV	R2,R3,R4
2	MUL	R1,R5,R6
3	ADD	R3,R7,R8
4	MUL	R1,R1,R3
5	SUB	R4,R1,R5
6	ADD	R1,R4,R2

Add= 1 Cycle
Mult= 10 Cycle
Div= 40 Cycle

RS Fields

Op	Dst-Tag	Tag 1	Tag 2	Val 1	Val 2

ROB Fields

Type	Dst	Value	Done	

RS Adder

RS Mul / Div

MUL	ROB2				3	4

Cycle # 2

R1	-23
R2	16
R3	45
R4	5
R5	3
R6	4
R7	1
R8	2

ARF

R1	ROB2
R2	ROB1
R3	
R4	
R5	
R6	
R7	
R8	

RAT

FUTURE VAL

ROB

ROB1	DIV	R2	45/5=9	
ROB2	MUL	R1		
ROB3				
ROB4				
ROB5				
ROB6				

Inst	Is	Ex	Wr	Cm
1	1	2	42	
2	2	3		
3				
4				
5				
6				

Sr	Inst	Operands
1	DIV	R2,R3,R4
2	MUL	R1,R5,R6
3	ADD	R3,R7,R8
4	MUL	R1,R1,R3
5	SUB	R4,R1,R5
6	ADD	R1,R4,R2

Add= 1 Cycle
Mult= 10 Cycle
Div= 40 Cycle

RS Fields

Op	Dst-Tag	Tag 1	Tag 2	Val 1	Val 2

ROB Fields

Type	Dst	Value	Done	

RS Adder

ADD	ROB3			1	2

RS Mul / Div

MUL	ROB2				3	4

Cycle # 3

ARF

R1	-23
R2	16
R3	45
R4	5
R5	3
R6	4
R7	1
R8	2

RAT

R1	ROB2
R2	ROB1
R3	ROB3
R4	
R5	
R6	
R7	
R8	

ROB

FUTURE VAL

ROB1	DIV	R2	45/5=9	
ROB2	MUL	R1		
ROB3	ADD	R3		
ROB4				
ROB5				
ROB6				

Inst	Is	Ex	Wr	Cm
1	1	2	42	
2	2	3	13	
3	3	4	5	
4				
5				
6				

Sr	Inst	Operands
	DIV	R2,R3,R4
	MUL	R1,R5,R6
	ADD	R3,R7,R8
	MUL	R1,R1,R3
	SUB	R4,R1,R5
	ADD	R1,R4,R2

Add= 1 Cycle
Mult= 10 Cycle
Div= 40 Cycle

RS Fields

Op	Dst-Tag	Tag 1	Tag 2	Val 1	Val 2

ROB Fields

Type	Dst	Value	Done	

RS Adder

ADD	ROB3			1	2

RS Mul / Div

MUL	ROB4	ROB2	ROB3			

Cycle # 4

R1	-23
R2	16
R3	45
R4	5
R5	3
R6	4
R7	1
R8	2

ARF

R1	ROB4
R2	ROB1
R3	ROB3
R4	
R5	
R6	
R7	
R8	

RAT

ROB

ROB1	DIV	R2		
ROB2	MUL	R1		
ROB3	ADD	R3		
ROB4	MUL	R1		
ROB5				
ROB6				

Inst	Is	Ex	Wr	Cm
1	1	2	42	
2	2	3	13	
3	3	4	5	
4	4			
5				
6				

Sr	Inst	Operands
1	DIV	R2,R3,R4
2	MUL	R1,R5,R6
3	ADD	R3,R7,R8
4	MUL	R1,R1,R3
5	SUB	R4,R1,R5
6	ADD	R1,R4,R2

Add= 1 Cycle
Mult= 10 Cycle
Div= 40 Cycle

RS Fields

Op	Dst-Tag	Tag 1	Tag 2	Val 1	Val 2

ROB Fields

Type	Dst	Value	Done	

RS Adder

SUB	ROB5	ROB4			3

RS Mul / Div

MUL	ROB4	ROB2				3

Cycle # 5

R1	-23
R2	16
R3	45
R4	5
R5	3
R6	4
R7	1
R8	2

ARF

R1	ROB4
R2	ROB1
R3	ROB3
R4	ROB5
R5	
R6	
R7	
R8	

RAT

ROB

ROB1	DIV	R2	FUT 9	
ROB2	MUL	R1	FUT 12	
ROB3	ADD	R3	3	YES
ROB4	MUL	R1		
ROB5	SUB	R4		
ROB6				

Inst	Is	Ex	Wr	Cm
1	1	2	42	
2	2	3	13	
3	3	4	5	
4	4			
5	5			
6				

Sr	Inst	Operands
1	DIV	R2,R3,R4
2	MUL	R1,R5,R6
3	ADD	R3,R7,R8
4	MUL	R1,R1,R3
5	SUB	R4,R1,R5
6	ADD	R1,R4,R2

Add= 1 Cycle
Mult= 10 Cycle
Div= 40 Cycle

RS Fields

Op	Dst-Tag	Tag 1	Tag 2	Val 1	Val 2

ROB Fields

Type	Dst	Value	Done	

RS Adder

SUB	ROB5	ROB4			3
ADD	ROB6	ROB5	ROB1		

RS Mul / Div

MUL	ROB4	ROB2				3

Cycle # 6

R1	-23
R2	16
R3	45
R4	5
R5	3
R6	4
R7	1
R8	2

ARF

R1	**ROB6**
R2	ROB1
R3	ROB3
R4	ROB5
R5	
R6	
R7	
R8	

RAT

ROB

ROB1	DIV	R2	FUT 9	
ROB2	MUL	R1	FUT 12	
ROB3	ADD	R3	3	YES
ROB4	MUL	R1		
ROB5	SUB	R4		
ROB6	ADD	R1		

Inst	Is	Ex	Wr	Cm
1	1	2	42	
2	2	3	13	
3	3	4	5	
4	4			
5	5			
6	6			

Sr	Inst	Operands
1	DIV	R2,R3,R4
2	MUL	R1,R5,R6
3	ADD	R3,R7,R8
4	MUL	R1,R1,R3
5	SUB	R4,R1,R5
6	ADD	R1,R4,R2

Add= 1 Cycle
Mult= 10 Cycle
Div= 40 Cycle

RS Fields	Op	Dst-Tag	Tag 1	Tag 2	Val 1	Val 2

ROB Fields	Type	Dst	Value	Done	

RS Adder

SUB	ROB5	ROB4			3
ADD	ROB6	ROB5	ROB1		

RS Mul / Div

MUL	ROB4	ROB2				3

Cycle #
7 -12
NO CHANGE

R1	-23
R2	16
R3	45
R4	5
R5	3
R6	4
R7	1
R8	2

ARF

R1	ROB6
R2	ROB1
R3	ROB3
R4	ROB5
R5	
R6	
R7	
R8	

RAT

ROB

ROB1	DIV	R2	FUT 9	
ROB2	MUL	R1	FUT 12	
ROB3	ADD	R3	3	YES
ROB4	MUL	R1		
ROB5	SUB	R4		
ROB6	ADD	R1		

Inst	Is	Ex	Wr	Cm
1	1	2	42	
2	2	3	13	
3	3	4	5	
4	4			
5	5			
6	6			

Sr	Inst	Operands
1	DIV	R2,R3,R4
2	MUL	R1,R5,R6
3	ADD	R3,R7,R8
4	MUL	R1,R1,R3
5	SUB	R4,R1,R5
6	ADD	R1,R4,R2

Add= 1 Cycle
Mult= 10 Cycle
Div= 40 Cycle

RS Fields	Op	Dst-Tag	Tag 1	Tag 2	Val 1	Val 2

ROB Fields	Type	Dst	Value	Done	

RS Adder

SUB	ROB5	ROB4			3
ADD	ROB6	ROB5	ROB1		

RS Mul / Div

MUL	ROB4				12	3

Cycle #
13

R1	-23
R2	16
R3	45
R4	5
R5	3
R6	4
R7	1
R8	2

ARF

R1	ROB6
R2	ROB1
R3	ROB3
R4	ROB5
R5	
R6	
R7	
R8	

RAT

ROB

ROB1	DIV	R2	FUT 9	
ROB2	MUL	R1	12	YES
ROB3	ADD	R3	3	YES
ROB4	MUL	R1		
ROB5	SUB	R4		
ROB6	ADD	R1		

Inst	Is	Ex	Wr	Cm
1	1	2	42	
2	2	3	13	
3	3	4	5	
4	4			
5	5			
6	6			

Sr	Inst	Operands
1	DIV	R2,R3,R4
2	MUL	R1,R5,R6
3	ADD	R3,R7,R8
4	MUL	R1,R1,R3
5	SUB	R4,R1,R5
6	ADD	R1,R4,R2

Add= 1 Cycle
Mult= 10 Cycle
Div= 40 Cycle

RS Fields

Op	Dst-Tag	Tag 1	Tag 2	Val 1	Val 2

ROB Fields

Type	Dst	Value	Done	

RS Adder

SUB	ROB5	ROB4			3
ADD	ROB6	ROB5	ROB1		

RS Mul / Div

MUL	ROB4				12	3

Cycle #
14 TO 23
SAME Status

R1	-23
R2	16
R3	45
R4	5
R5	3
R6	4
R7	1
R8	2

ARF

R1	ROB6
R2	ROB1
R3	ROB3
R4	ROB5
R5	
R6	
R7	
R8	

RAT

ROB

ROB1	DIV	R2	FUT 9	
ROB2	MUL	R1	12	YES
ROB3	ADD	R3	3	YES
ROB4	MUL	R1	FUT 36	
ROB5	SUB	R4		
ROB6	ADD	R1		

Inst	Is	Ex	Wr	Cm
1	1	2	42	
2	2	3	13	
3	3	4	5	
4	4	14	24	
5	5			
6	6			

Sr	Inst	Operands
1	DIV	R2,R3,R4
2	MUL	R1,R5,R6
3	ADD	R3,R7,R8
4	MUL	R1,R1,R3
5	SUB	R4,R1,R5
6	ADD	R1,R4,R2

Add= 1 Cycle
Mult= 10 Cycle
Div= 40 Cycle

RS Fields

Op	Dst-Tag	Tag 1	Tag 2	Val 1	Val 2

ROB Fields

Type	Dst	Value	Done	

RS Adder

SUB	ROB5			36	3
ADD	ROB6	ROB5	ROB1		

RS Mul / Div

MUL	ROB4				12	3

Cycle #
24

ARF

R1	-23
R2	16
R3	45
R4	5
R5	3
R6	4
R7	1
R8	2

RAT

R1	ROB6
R2	ROB1
R3	ROB3
R4	ROB5
R5	
R6	
R7	
R8	

ROB

ROB1	DIV	R2	FUT 9	
ROB2	MUL	R1	12	YES
ROB3	ADD	R3	3	YES
ROB4	MUL	R1	36	YES
ROB5	SUB	R4		
ROB6	ADD	R1		

Inst	Is	Ex	Wr	Cm
1	1	2	42	
2	2	3	13	
3	3	4	5	
4	4	14	24	
5	5			
6	6			

Sr	Inst	Operands
1	DIV	R2,R3,R4
2	MUL	R1,R5,R6
3	ADD	R3,R7,R8
4	MUL	R1,R1,R3
5	SUB	R4,R1,R5
6	ADD	R1,R4,R2

Add= 1 Cycle
Mult= 10 Cycle
Div= 40 Cycle

RS Fields

Op	Dst-Tag	Tag 1	Tag 2	Val 1	Val 2

ROB Fields

Type	Dst	Value	Done	

RS Mul / Div

MUL	ROB4				12	3

RS Adder

SUB	ROB5			36	3
ADD	ROB6	ROB5	ROB1		

Cycle #
25

R1	-23
R2	16
R3	45
R4	5
R5	3
R6	4
R7	1
R8	2

ARF

R1	ROB6
R2	ROB1
R3	ROB3
R4	ROB5
R5	
R6	
R7	
R8	

RAT

ROB

ROB1	DIV	R2	FUT 9	
ROB2	MUL	R1	12	YES
ROB3	ADD	R3	3	YES
ROB4	MUL	R1	36	YES
ROB5	SUB	R4	FUT 33	
ROB6	ADD	R1		

Inst	Is	Ex	Wr	Cm
1	1	2	42	
2	2	3	13	
3	3	4	5	
4	4	14	24	
5	5	25	26	
6	6			

Sr	Inst	Operands
1	DIV	R2,R3,R4
2	MUL	R1,R5,R6
3	ADD	R3,R7,R8
4	MUL	R1,R1,R3
5	SUB	R4,R1,R5
6	ADD	R1,R4,R2

Add= 1 Cycle
Mult= 10 Cycle
Div= 40 Cycle

RS Fields

Op	Dst-Tag	Tag 1	Tag 2	Val 1	Val 2

ROB Fields

Type	Dst	Value	Done	

RS Adder

SUB	ROB5			36	3
ADD	ROB6		ROB1	33	

RS Mul / Div

MUL	ROB4				12	3

Cycle #
26 TO 41
SAME Status

ARF

R1	-23
R2	16
R3	45
R4	5
R5	3
R6	4
R7	1
R8	2

RAT

R1	ROB6
R2	ROB1
R3	ROB3
R4	ROB5
R5	
R6	
R7	
R8	

ROB

ROB1	DIV	R2	FUT 9	
ROB2	MUL	R1	12	YES
ROB3	ADD	R3	3	YES
ROB4	MUL	R1	36	YES
ROB5	SUB	R4	33	YES
ROB6	ADD	R1		

Inst	Is	Ex	Wr	Cm
1	1	2	42	
2	2	3	13	
3	3	4	5	
4	4	14	24	
5	5	25	26	
6	6			

Sr	Inst	Operands
1	DIV	R2,R3,R4
2	MUL	R1,R5,R6
3	ADD	R3,R7,R8
4	MUL	R1,R1,R3
5	SUB	R4,R1,R5
6	ADD	R1,R4,R2

Add= 1 Cycle
Mult= 10 Cycle
Div= 40 Cycle

RS Fields

Op	Dst-Tag	Tag 1	Tag 2	Val 1	Val 2

ROB Fields

Type	Dst	Value	Done	

RS Mul / Div

MUL	ROB4				12	3

RS Adder

ADD	ROB6		9	33	

Cycle #
42

R1	-23
R2	16
R3	45
R4	5
R5	3
R6	4
R7	1
R8	2

ARF

R1	ROB6
R2	ROB1
R3	ROB3
R4	ROB5
R5	
R6	
R7	
R8	

RAT

ROB

ROB1	DIV	R2	9	YES
ROB2	MUL	R1	12	YES
ROB3	ADD	R3	3	YES
ROB4	MUL	R1	36	YES
ROB5	SUB	R4	33	YES
ROB6	ADD	R1		

Inst	Is	Ex	Wr	Cm
1	1	2	42	
2	2	3	13	
3	3	4	5	
4	4	14	24	
5	5	25	26	
6	6			

Sr	Inst	Operands	
1	DIV	R2,R3,R4	YES
2	MUL	R1,R5,R6	
3	ADD	R3,R7,R8	
4	MUL	R1,R1,R3	
5	SUB	R4,R1,R5	
6	ADD	R1,R4,R2	

Add= 1 Cycle
Mult= 10 Cycle
Div= 40 Cycle

RS Fields

Op	Dst-Tag	Tag 1	Tag 2	Val 1	Val 2

ROB Fields

Type	Dst	Value	Done	

RS Mul / Div

MUL	ROB4				12	3

RS Adder

~~ADD~~	~~ROB6~~		~~9~~	~~33~~	

Cycle #
43

ARF

R1	-23
R2	~~16~~ **TO 9**
R3	45
R4	5
R5	3
R6	4
R7	1
R8	2

RAT

R1	ROB6
R2	~~ROB1~~
R3	ROB3
R4	ROB5
R5	
R6	
R7	
R8	

ROB

~~ROB1~~	~~DIV~~	~~R2~~	~~9~~	~~YES~~
ROB2	MUL	R1	12	YES
ROB3	ADD	R3	3	YES
ROB4	MUL	R1	36	YES
ROB5	SUB	R4	33	YES
ROB6	ADD	R1	FUT 42	

Inst	Is	Ex	Wr	Cm
1	1	2	42	43
2	2	3	13	
3	3	4	5	
4	4	14	24	
5	5	25	26	
6	6	43	44	

Sr	Inst	Operands	
1	DIV	R2,R3,R4	YES
2	MUL	R1,R5,R6	YES
3	ADD	R3,R7,R8	
4	MUL	R1,R1,R3	
5	SUB	R4,R1,R5	
6	ADD	R1,R4,R2	

Add= 1 Cycle
Mult= 10 Cycle
Div= 40 Cycle

RS Fields

Op	Dst-Tag	Tag 1	Tag 2	Val 1	Val 2

ROB Fields

Type	Dst	Value	Done	

RS Adder

RS Mul / Div

MUL	ROB4				12	3

Cycle #
44

ARF

R1	~~23~~ **12**
R2	**9**
R3	45
R4	5
R5	3
R6	4
R7	1
R8	2

RAT

R1	ROB6
R2	
R3	ROB3
R4	ROB5
R5	
R6	
R7	
R8	

ROB

~~ROB2~~	~~MUL~~	~~R1~~	~~12~~	~~YES~~
ROB3	ADD	R3	3	YES
ROB4	MUL	R1	36	YES
ROB5	SUB	R4	33	YES
ROB6	ADD	R1	42	YES

Inst	Is	Ex	Wr	Cm
1	1	2	42	43
2	2	3	13	44
3	3	4	5	
4	4	14	24	
5	5	25	26	
6	6	43	44	

Sr	Inst	Operands	
1	DIV	R2,R3,R4	YES
2	MUL	R1,R5,R6	YES
3	ADD	R3,R7,R8	
4	MUL	R1,R1,R3	
5	SUB	R4,R1,R5	
6	ADD	R1,R4,R2	

Add= 1 Cycle
Mult= 10 Cycle
Div= 40 Cycle

RS Fields

Op	Dst-Tag	Tag 1	Tag 2	Val 1	Val 2

ROB Fields

Type	Dst	Value	Done	

RS Adder

RS Mul / Div

MUL	ROB4				12	3

Cycle #
45

ARF

R1	**12**
R2	**9**
R3	~~45~~ **3**
R4	5
R5	3
R6	4
R7	1
R8	2

RAT

R1	ROB6
R2	
R3	~~ROB3~~
R4	ROB5
R5	
R6	
R7	
R8	

ROB

~~ROB3~~	~~ADD~~	~~R3~~	~~3~~	~~YES~~
ROB4	MUL	R1	36	YES
ROB5	SUB	R4	33	YES
ROB6	ADD	R1	42	YES

Inst	Is	Ex	Wr	Cm
1	1	2	42	43
2	2	3	13	44
3	3	4	5	45
4	4	14	24	
5	5	25	26	
6	6	43	44	

Sr	Inst	Operands	
1	DIV	R2,R3,R4	YES
2	MUL	R1,R5,R6	YES
3	ADD	R3,R7,R8	
4	MUL	R1,R1,R3	
5	SUB	R4,R1,R5	
6	ADD	R1,R4,R2	

Add= 1 Cycle
Mult= 10 Cycle
Div= 40 Cycle

RS Fields

Op	Dst-Tag	Tag 1	Tag 2	Val 1	Val 2

ROB Fields

Type	Dst	Value	Done	

RS Adder

RS Mul / Div

MUL	ROB4				12	3

Cycle #
46

ARF

R1	**12**
R2	**9**
R3	**3**
R4	5
R5	3
R6	4
R7	1
R8	2

RAT

R1	ROB6
R2	
R3	
R4	ROB5
R5	
R6	
R7	
R8	

ROB

~~ROB4~~	~~MUL~~	~~R1~~	~~36~~	~~YES~~
ROB5	SUB	R4	33	YES
ROB6	ADD	R1	42	YES

Inst	Is	Ex	Wr	Cm
1	1	2	42	43
2	2	3	13	44
3	3	4	5	45
4	4	14	24	46
5	5	25	26	
6	6	43	44	

Sr	Inst	Operands	
1	DIV	R2,R3,R4	YES
2	MUL	R1,R5,R6	YES
3	ADD	R3,R7,R8	
4	MUL	R1,R1,R3	
5	SUB	R4,R1,R5	
6	ADD	R1,R4,R2	

Add= 1 Cycle
Mult= 10 Cycle
Div= 40 Cycle

RS Fields

Op	Dst-Tag	Tag 1	Tag 2	Val 1	Val 2

ROB Fields

Type	Dst	Value	Done	

RS Adder

RS Mul / Div

MUL	ROB4				12	3

Cycle #
47

ARF

R1	**12**
R2	**9**
R3	**3**
R4	~~5~~ **33**
R5	3
R6	4
R7	1
R8	2

RAT

R1	ROB6
R2	
R3	
R4	~~ROB5~~
R5	
R6	
R7	
R8	

ROB

~~ROB5~~	~~SUB~~	~~R4~~	~~33~~	~~YES~~
ROB6	ADD	R1	42	YES

Inst	Is	Ex	Wr	Cm
1	1	2	42	43
2	2	3	13	44
3	3	4	5	45
4	4	14	24	46
5	5	25	26	47
6	6	43	44	

Sr	Inst	Operands	
1	DIV	R2,R3,R4	YES
2	MUL	R1,R5,R6	YES
3	ADD	R3,R7,R8	YES
4	MUL	R1,R1,R3	YES
5	SUB	R4,R1,R5	YES
6	ADD	R1,R4,R2	YES

Add= 1 Cycle
Mult= 10 Cycle
Div= 40 Cycle

RS Fields

Op	Dst-Tag	Tag 1	Tag 2	Val 1	Val 2

ROB Fields

Type	Dst	Value	Done	

RS Adder

RS Mul / Div

MUL	ROB4				12	3

Cycle #
48

ARF

R1	~~12~~ **42**
R2	**9**
R3	**3**
R4	**33**
R5	3
R6	4
R7	1
R8	2

RAT

R1	~~ROB6~~
R2	
R3	
R4	ROB5
R5	
R6	
R7	
R8	

ROB

ROB6	ADD	R1	42	YES

Inst	Is	Ex	Wr	Cm
1	1	2	42	43
2	2	3	13	44
3	3	4	5	45
4	4	14	24	46
5	5	25	26	47
6	6	43	44	48

Sr	Inst	Operands	
1	DIV	R2,R3,R4	YES
2	MUL	R1,R5,R6	YES
3	ADD	R3,R7,R8	YES
4	MUL	R1,R1,R3	YES
5	SUB	R4,R1,R5	YES
6	ADD	R1,R4,R2	YES

Add= 1 Cycle
Mult= 10 Cycle
Div= 40 Cycle

RS Fields	Op	Dst-Tag	Tag 1	Tag 2	Val 1	Val 2

ROB Fields	Type	Dst	Value	Done	

RS Adder

RS Mul / Div

MUL	ROB4				12	3

Cycle #
49
ALL
COMMITTED

R1	**42**
R2	**9**
R3	**3**
R4	**33**
R5	3
R6	4
R7	1
R8	2

ARF

R1	
R2	
R3	
R4	ROB5
R5	
R6	
R7	
R8	

RAT

ROB

Inst	Is	Ex	Wr	Cm
1	1	2	42	43
2	2	3	13	44
3	3	4	5	45
4	4	14	24	46
5	5	25	26	47
6	6	43	44	48